City Liveability & Urban Climate Insights API

Version: 1.0.0

Aggregates urban climate, air quality, and socioeconomic data for UK cities. Computes composite liveability scores and provides a natural-language query interface.

Authentication

All protected endpoints require a Bearer token obtained from `POST /api/v1/auth/login`.

Include the token in the `Authorization` header:

```
Authorization: Bearer <access_token>
```

Analytics

`GET /api/v1/cities/{city_id}/liveability`

Get Liveability

Parameters:

Name	In	Type	Required	Description
`city_id`	path	integer	Yes	

Responses:

Status	Description
200	Successful Response
422	Validation Error

Example Request:

```
curl http://localhost:8000/api/v1/cities/1/liveability
```

Example Response (200):

```
{
  "city_id": 1,
  "city_name": "Leeds",
  "overall_score": 61.4,
  "climate_score": 52.3,
  "affordability_score": 68.0,
```

```
  "safety_score": 72.0,
  "environment_score": 55.0,
  "data_points": 365
}
```

GET /api/v1/cities/compare

Compare Cities

Parameters:

Name	In	Type	Required	Description
ids	query	string	Yes	Comma-separated city IDs

Responses:

Status	Description
200	Successful Response
422	Validation Error

Example Request:

```
curl "http://localhost:8000/api/v1/cities/compare?ids=1,2,3"
```

Example Response (200):


```
[
  {
    "city_id": 1,
    "city_name": "Leeds",
    "overall_score": 61.4,
    "climate_score": 52.3,
    "affordability_score": 68.0,
    "safety_score": 72.0,
    "environment_score": 55.0,
    "data_points": 365
  },
  {
    "city_id": 2,
    "city_name": "Manchester",
    "overall_score": 58.7,
    "climate_score": 49.1,
    "affordability_score": 64.5,
    "safety_score": 65.0,
```

```
    "environment_score": 58.0,
    "data_points": 365
  },
  {
    "city_id": 3,
    "city_name": "Bristol",
    "overall_score": 66.2,
    "climate_score": 60.8,
    "affordability_score": 52.0,
    "safety_score": 78.0,
    "environment_score": 72.0,
    "data_points": 365
  }
]
```

`GET /api/v1/cities/{city_id}/trends`

Get Trends

Parameters:

Name	In	Type	Required	Description
`city_id`	path	integer	Yes	
`metric`	query	string	No	Metric to trend (`avg_temp_c`, `aqi`, `humidity_pct`, `precipitation_mm`)
`period`	query	string	No	Period e.g. `6m`, `12m`, `24m`

Responses:

Status	Description
200	Successful Response
422	Validation Error

Example Request:

```
curl "http://localhost:8000/api/v1/cities/1/trends?metric=avg_temp_c&period=6m"
```

Example Response (200):


```
{
  "city_id": 1,
```

```
  "city_name": "Leeds",
  "metric": "avg_temp_c",
  "period": "6m",
  "data_points": [
    {"date": "2025-09-01", "value": 16.2},
    {"date": "2025-10-01", "value": 12.4},
    {"date": "2025-11-01", "value": 8.7},
    {"date": "2025-12-01", "value": 5.1},
    {"date": "2026-01-01", "value": 3.8},
    {"date": "2026-02-01", "value": 4.9}
  ]
}
```

`GET /api/v1/cities/{city_id}/anomalies`

Get Anomalies

Parameters:

Name	In	Type	Required	Description
`city_id`	path	integer	Yes	
`threshold`	query	number	No	Z-score threshold (default: 2.0)

Responses:

Status	Description
200	Successful Response
422	Validation Error

Example Request:

```
curl "http://localhost:8000/api/v1/cities/1/anomalies?threshold=2.0"
```

Example Response (200):


```
{
  "city_id": 1,
  "city_name": "Leeds",
  "threshold": 2.0,
  "anomalies": [
    {
      "date": "2025-07-19",
      "metric": "avg_temp_c",
      "value": 31.4,
```

```
      "z_score": 3.12
    },
    {
      "date": "2026-01-08",
      "metric": "precipitation_mm",
      "value": 42.1,
      "z_score": 2.87
    }
  ]
}
```

Authentication

`POST /api/v1/auth/register`

Register

Request Body (`application/json`): `UserRegister`

Responses:

Status	Description
201	Successful Response
422	Validation Error

Example Request:

```
curl -X POST http://localhost:8000/api/v1/auth/register \
  -H "Content-Type: application/json" \
  -d '{
    "username": "alice",
    "email": "alice@example.com",
    "password": "securepass123"
  }'
```

Example Response (201):

```
{
  "id": 1,
  "username": "alice",
  "email": "alice@example.com",
  "role": "user",
  "created_at": "2026-03-20T10:00:00"
}
```

Example Response (422 — validation error):

```
{
  "detail": [
    {
      "loc": ["body", "email"],
      "msg": "value is not a valid email address",
      "type": "value_error.email"
    }
  ]
}
```

`POST /api/v1/auth/login`

Login

Request Body (`application/json`): `UserLogin`

Responses:

Status	Description
200	Successful Response
422	Validation Error

Example Request:

```
curl -X POST http://localhost:8000/api/v1/auth/login \
  -H "Content-Type: application/json" \
  -d '{
    "username": "alice",
    "password": "securepass123"
  }'
```

Example Response (200):


```
{
  "access_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiJhbGljZSIsImV4cCI6MTc0MjQ3
  "token_type": "bearer"
}
```


Example Response (401 — invalid credentials):

```
{
  "detail": "Invalid username or password",
  "code": "INVALID_CREDENTIALS"
}
```

`GET /api/v1/auth/me`

Get Me

Responses:

Status	Description
200	Successful Response

Example Request:

```
curl http://localhost:8000/api/v1/auth/me \
  -H "Authorization: Bearer <access_token>"
```

Example Response (200):

```
{
  "id": 1,
  "username": "alice",
  "email": "alice@example.com",
  "role": "user",
  "created_at": "2026-03-20T10:00:00"
}
```

`POST /api/v1/auth/refresh`

Refresh Token

Issue a new access token for an authenticated user.

The caller must present a valid (non-expired) Bearer token. A fresh token is returned with a reset expiry window.

Responses:

Status	Description
200	Successful Response

Example Request:

```
curl -X POST http://localhost:8000/api/v1/auth/refresh \
  -H "Authorization: Bearer <access_token>"
```

Example Response (200):

```
{
  "access_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiJhbGljZSIsImV4cCI6MTc0MjQ3
  "token_type": "bearer"
}
```

Cities

`POST /api/v1/cities`

Create City

Create a new city. Automatically triggers a background fetch of the last 365 days of weather data from the Open-Meteo archive API, stored as climate metrics.

Request Body (`application/json`): `CityCreate`

Responses:

Status	Description
201	Successful Response
422	Validation Error

Example Request:

```
curl -X POST http://localhost:8000/api/v1/cities \
  -H "Content-Type: application/json" \
  -H "Authorization: Bearer <access_token>" \
  -d '{
    "name": "Leeds",
    "region": "Yorkshire",
    "country": "UK",
    "latitude": 53.8008,
    "longitude": -1.5491,
    "population": 812000
  }'
```

Example Response (201):


```
{
  "id": 1,
  "name": "Leeds",
  "region": "Yorkshire",
  "country": "UK",
  "latitude": 53.8008,
  "longitude": -1.5491,
  "population": 812000,
```

```
  "created_at": "2026-03-20T10:05:00",
  "updated_at": "2026-03-20T10:05:00"
}
```

`GET /api/v1/cities`

List Cities

Parameters:

Name	In	Type	Required	Description
`offset`	query	integer	No	Pagination offset (default: 0)
`limit`	query	integer	No	Results per page, max 100 (default: 20)
`region`	query	string	No	Filter by region name
`sort`	query	string	No	Sort field, prefix with `-` for descending (e.g. `-population`)

Responses:

Status	Description
200	Successful Response
422	Validation Error

Example Request:

```
curl "http://localhost:8000/api/v1/cities?region=Yorkshire&limit=5&sort=-population"
```

Example Response (200):

```
{
  "items": [
    {
      "id": 1,
      "name": "Leeds",
      "region": "Yorkshire",
      "country": "UK",
      "latitude": 53.8008,
      "longitude": -1.5491,
      "population": 812000,
      "created_at": "2026-03-20T10:05:00",
      "updated_at": "2026-03-20T10:05:00"
```

```
    },
    {
      "id": 4,
      "name": "Sheffield",
      "region": "Yorkshire",
      "country": "UK",
      "latitude": 53.3811,
      "longitude": -1.4701,
      "population": 584000,
      "created_at": "2026-03-20T10:06:00",
      "updated_at": "2026-03-20T10:06:00"
    }
  ],
  "total": 2,
  "offset": 0,
  "limit": 5
}
```

`GET /api/v1/cities/{city_id}`

Get City

Parameters:

Name	In	Type	Required	Description
`city_id`	path	integer	Yes	

Responses:

Status	Description
200	Successful Response
422	Validation Error

Example Request:

```
curl http://localhost:8000/api/v1/cities/1
```

Example Response (200):


```
{
  "id": 1,
  "name": "Leeds",
  "region": "Yorkshire",
  "country": "UK",
```

```
  "latitude": 53.8008,
  "longitude": -1.5491,
  "population": 812000,
  "created_at": "2026-03-20T10:05:00",
  "updated_at": "2026-03-20T10:05:00"
}
```

Example Response (404 — not found):

```
{
  "detail": "City not found",
  "code": "NOT_FOUND"
}
```

`PUT /api/v1/cities/{city_id}`

Update City

Update a city's details. Re-triggers a background fetch of the last 365 days of weather data from the Open-Meteo archive API; duplicate dates are skipped automatically.

Parameters:

Name	In	Type	Required	Description
`city_id`	path	integer	Yes	

Request Body (`application/json`): `CityUpdate`

Responses:

Status	Description
200	Successful Response
422	Validation Error

Example Request:

```
curl -X PUT http://localhost:8000/api/v1/cities/1 \
  -H "Content-Type: application/json" \
  -H "Authorization: Bearer <access_token>" \
  -d '{
    "population": 830000
  }'
```

Example Response (200):

```
{
  "id": 1,
  "name": "Leeds",
  "region": "Yorkshire",
  "country": "UK",
  "latitude": 53.8008,
  "longitude": -1.5491,
  "population": 830000,
  "created_at": "2026-03-20T10:05:00",
  "updated_at": "2026-03-20T11:00:00"
}
```

`DELETE /api/v1/cities/{city_id}`

Delete City *(Admin only)*

Parameters:

Name	In	Type	Required	Description
`city_id`	path	integer	Yes	

Responses:

Status	Description
204	Successful Response (no body)
403	Forbidden — admin role required
422	Validation Error

Example Request:

```
curl -X DELETE http://localhost:8000/api/v1/cities/1 \
  -H "Authorization: Bearer <admin_access_token>"
```

Example Response (204): *(empty body)*

Example Response (403 — insufficient permissions):

```
{
  "detail": "Admin access required",
  "code": "FORBIDDEN"
}
```

City Narrative

`GET /api/v1/cities/{city_id}/narrative`

Get City Narrative

Returns a textual summary of a city's liveability based on its computed scores.

Parameters:

Name	In	Type	Required	Description
`city_id`	path	integer	Yes	

Responses:

Status	Description
200	Successful Response
422	Validation Error

Example Request:

```
curl http://localhost:8000/api/v1/cities/1/narrative
```

Example Response (200):


```
{
  "city_id": 1,
  "city_name": "Leeds",
  "narrative": "Leeds has an overall liveability score of 61.4/100. Climate scores 52.3/100,
}
```

Health

`GET /`

Health Check

Responses:

Status	Description
200	Successful Response

Example Request:

```
curl http://localhost:8000/
```

Example Response (200):

```
{
  "status": "ok",
  "version": "1.0.0"
}
```

Metrics

`POST /api/v1/cities/{city_id}/climate-metrics`

Create Climate Metric

Parameters:

Name	In	Type	Required	Description
`city_id`	path	integer	Yes	

Request Body (`application/json`): `ClimateMetricCreate`

Responses:

Status	Description
201	Successful Response
422	Validation Error

Example Request:

```
curl -X POST http://localhost:8000/api/v1/cities/1/climate-metrics \
  -H "Content-Type: application/json" \
  -H "Authorization: Bearer <access_token>" \
  -d '{
    "date": "2026-03-20",
    "avg_temp_c": 8.5,
    "aqi": 42.0,
    "humidity_pct": 78.0,
    "precipitation_mm": 3.2,
    "source": "manual"
  }'
```

Example Response (201):

```
{
  "id": 1001,
  "city_id": 1,
```

```
  "date": "2026-03-20",
  "avg_temp_c": 8.5,
  "aqi": 42.0,
  "humidity_pct": 78.0,
  "precipitation_mm": 3.2,
  "source": "manual",
  "created_at": "2026-03-20T12:00:00"
}
```

`GET /api/v1/cities/{city_id}/climate-metrics`

List Climate Metrics

Parameters:

Name	In	Type	Required	Description
`city_id`	path	integer	Yes	
`start_date`	query	string	No	ISO date, e.g. `2026-01-01`
`end_date`	query	string	No	ISO date, e.g. `2026-03-20`
`offset`	query	integer	No	Pagination offset (default: 0)
`limit`	query	integer	No	Results per page, max 100 (default: 20)

Responses:

Status	Description
200	Successful Response
422	Validation Error

Example Request:

```
curl "http://localhost:8000/api/v1/cities/1/climate-metrics?start_date=2026-03-01&end_date=2
```

Example Response (200):


```
{
  "items": [
    {
      "id": 999,
      "city_id": 1,
      "date": "2026-03-01",
      "avg_temp_c": 7.1,
      "aqi": 38.5,
      "humidity_pct": 81.0,
      "precipitation_mm": 1.4,
```

```
      "source": "open_meteo",
      "created_at": "2026-03-20T10:05:12"
    },
    {
      "id": 1000,
      "city_id": 1,
      "date": "2026-03-10",
      "avg_temp_c": 9.3,
      "aqi": 35.0,
      "humidity_pct": 74.5,
      "precipitation_mm": 0.0,
      "source": "open_meteo",
      "created_at": "2026-03-20T10:05:12"
    },
    {
      "id": 1001,
      "city_id": 1,
      "date": "2026-03-20",
      "avg_temp_c": 8.5,
      "aqi": 42.0,
      "humidity_pct": 78.0,
      "precipitation_mm": 3.2,
      "source": "manual",
      "created_at": "2026-03-20T12:00:00"
    }
  ],
  "total": 3,
  "offset": 0,
  "limit": 3
}
```

`GET /api/v1/climate-metrics/{metric_id}`

Get Climate Metric

Parameters:

Name	In	Type	Required	Description
`metric_id`	path	integer	Yes	

Responses:

Status	Description
200	Successful Response
422	Validation Error

Example Request:

```
curl http://localhost:8000/api/v1/climate-metrics/1001
```

Example Response (200):

```
{
  "id": 1001,
  "city_id": 1,
  "date": "2026-03-20",
  "avg_temp_c": 8.5,
  "aqi": 42.0,
  "humidity_pct": 78.0,
  "precipitation_mm": 3.2,
  "source": "manual",
  "created_at": "2026-03-20T12:00:00"
}
```

`DELETE /api/v1/climate-metrics/{metric_id}`

Delete Climate Metric *(Admin only)*

Parameters:

Name	In	Type	Required	Description
`metric_id`	path	integer	Yes	

Responses:

Status	Description
204	Successful Response (no body)
403	Forbidden — admin role required
422	Validation Error

Example Request:

```
curl -X DELETE http://localhost:8000/api/v1/climate-metrics/1001 \
  -H "Authorization: Bearer <admin_access_token>"
```

Example Response (204): *(empty body)*

`POST /api/v1/cities/{city_id}/socioeconomic-metrics`

Create Socioeconomic Metric

Parameters:

Name	In	Type	Required	Description
`city_id`	path	integer	Yes	

Request Body (`application/json`): `SocioeconomicMetricCreate`

Responses:

Status	Description
201	Successful Response
422	Validation Error

Example Request:

```
curl -X POST http://localhost:8000/api/v1/cities/1/socioeconomic-metrics \
  -H "Content-Type: application/json" \
  -H "Authorization: Bearer <access_token>" \
  -d '{
    "year": 2025,
    "median_rent_gbp": 950,
    "green_space_pct": 22.5,
    "crime_index": 45.0,
    "avg_commute_min": 34,
    "source": "ONS"
  }'
```

Example Response (201):

```
{
  "id": 42,
  "city_id": 1,
  "year": 2025,
  "median_rent_gbp": 950,
  "green_space_pct": 22.5,
  "crime_index": 45.0,
  "avg_commute_min": 34,
  "source": "ONS",
```

```
  "created_at": "2026-03-20T12:05:00"
}
```

`GET /api/v1/cities/{city_id}/socioeconomic-metrics`

List Socioeconomic Metrics

Parameters:

Name	In	Type	Required	Description
`city_id`	path	integer	Yes	
`offset`	query	integer	No	Pagination offset (default: 0)
`limit`	query	integer	No	Results per page, max 100 (default: 20)

Responses:

Status	Description
200	Successful Response
422	Validation Error

Example Request:

```
curl "http://localhost:8000/api/v1/cities/1/socioeconomic-metrics"
```

Example Response (200):


```
{
  "items": [
    {
      "id": 40,
      "city_id": 1,
      "year": 2023,
      "median_rent_gbp": 875,
      "green_space_pct": 22.5,
      "crime_index": 47.0,
      "avg_commute_min": 36,
      "source": "ONS",
      "created_at": "2026-03-20T10:05:00"
    },
    {
      "id": 42,
      "city_id": 1,
      "year": 2025,
      "median_rent_gbp": 950,
```

```
      "green_space_pct": 22.5,
      "crime_index": 45.0,
      "avg_commute_min": 34,
      "source": "ONS",
      "created_at": "2026-03-20T12:05:00"
    }
  ],
  "total": 2,
  "offset": 0,
  "limit": 20
}
```

GET /api/v1/socioeconomic-metrics/{metric_id}

Get Socioeconomic Metric

Parameters:

Name	In	Type	Required	Description
`metric_id`	path	integer	Yes	

Responses:

Status	Description
200	Successful Response
422	Validation Error

Example Request:

```
curl http://localhost:8000/api/v1/socioeconomic-metrics/42
```

Example Response (200):


```
{
  "id": 42,
  "city_id": 1,
  "year": 2025,
  "median_rent_gbp": 950,
  "green_space_pct": 22.5,
  "crime_index": 45.0,
  "avg_commute_min": 34,
  "source": "ONS",
  "created_at": "2026-03-20T12:05:00"
}
```

`DELETE /api/v1/socioeconomic-metrics/{metric_id}`

Delete Socioeconomic Metric *(Admin only)*

Parameters:

Name	In	Type	Required	Description
`metric_id`	path	integer	Yes	

Responses:

Status	Description
204	Successful Response (no body)
403	Forbidden — admin role required
422	Validation Error

Example Request:

```
curl -X DELETE http://localhost:8000/api/v1/socioeconomic-metrics/42 \
  -H "Authorization: Bearer <admin_access_token>"
```

Example Response (204): *(empty body)*

Observations

`POST /api/v1/cities/{city_id}/observations`

Create Observation

Parameters:

Name	In	Type	Required	Description
`city_id`	path	integer	Yes	

Request Body (`application/json`): `ObservationCreate`

Responses:

Status	Description
201	Successful Response

Status	Description
422	Validation Error

Example Request:

```
curl -X POST http://localhost:8000/api/v1/cities/1/observations \
  -H "Content-Type: application/json" \
  -H "Authorization: Bearer <access_token>" \
  -d '{
    "category": "air_quality",
    "value": 38.5,
    "note": "Noticeably cleaner after the rain"
  }'
```

Example Response (201):

```
{
  "id": 15,
  "city_id": 1,
  "user_id": 1,
  "category": "air_quality",
  "value": 38.5,
  "note": "Noticeably cleaner after the rain",
  "recorded_at": "2026-03-20T14:30:00",
  "created_at": "2026-03-20T14:30:00"
}
```

`GET /api/v1/cities/{city_id}/observations`

List Observations

Parameters:

Name	In	Type	Required	Description
`city_id`	path	integer	Yes	
`offset`	query	integer	No	Pagination offset (default: 0)
`limit`	query	integer	No	Results per page, max 100 (default: 20)

Responses:

Status	Description
200	Successful Response

Status	Description
422	Validation Error

Example Request:

```
curl "http://localhost:8000/api/v1/cities/1/observations?limit=2"
```

Example Response (200):


```
{
  "items": [
    {
      "id": 14,
      "city_id": 1,
      "user_id": 1,
      "category": "noise_level",
      "value": 65.0,
      "note": "Heavy traffic near the ring road",
      "recorded_at": "2026-03-19T09:00:00",
      "created_at": "2026-03-19T09:00:00"
    },
    {
      "id": 15,
      "city_id": 1,
      "user_id": 1,
      "category": "air_quality",
      "value": 38.5,
      "note": "Noticeably cleaner after the rain",
      "recorded_at": "2026-03-20T14:30:00",
      "created_at": "2026-03-20T14:30:00"
    }
  ],
  "total": 2,
  "offset": 0,
  "limit": 2
}
```

`PUT /api/v1/observations/{obs_id}`

Update Observation *(Owner only)*

Parameters:

Name	In	Type	Required	Description
`obs_id`	path	integer	Yes	

Request Body (`application/json`): `ObservationUpdate`

Responses:

Status	Description
200	Successful Response
403	Forbidden — must be observation owner
422	Validation Error

Example Request:

```
curl -X PUT http://localhost:8000/api/v1/observations/15 \
  -H "Content-Type: application/json" \
  -H "Authorization: Bearer <access_token>" \
  -d '{
    "value": 35.0,
    "note": "Updated after re-reading the sensor"
  }'
```

Example Response (200):

```
{
  "id": 15,
  "city_id": 1,
  "user_id": 1,
  "category": "air_quality",
  "value": 35.0,
  "note": "Updated after re-reading the sensor",
  "recorded_at": "2026-03-20T14:30:00",
  "created_at": "2026-03-20T14:30:00"
}
```

`DELETE /api/v1/observations/{obs_id}`

Delete Observation *(Owner only)*

Parameters:

Name	In	Type	Required	Description
`obs_id`	path	integer	Yes	

Responses:

Status	Description
204	Successful Response (no body)
403	Forbidden — must be observation owner
422	Validation Error

Example Request:

```
curl -X DELETE http://localhost:8000/api/v1/observations/15 \
  -H "Authorization: Bearer <access_token>"
```

Example Response (204): *(empty body)*

Data Models

CityCreate

Field	Type	Required
`name`	string	Yes
`region`	string	Yes
`country`	string	No
`latitude`	number	Yes
`longitude`	number	Yes
`population`	integer	No

CityResponse

Field	Type	Required
`id`	integer	Yes
`name`	string	Yes
`region`	string	Yes
`country`	string	Yes
`latitude`	number	Yes
`longitude`	number	Yes
`population`	integer	Yes
`created_at`	string (datetime)	No

Field	Type	Required
updated_at	string (datetime)	No

CityUpdate

Field	Type	Required
name	string	No
region	string	No
country	string	No
latitude	number	No
longitude	number	No
population	integer	No

ClimateMetricCreate

Field	Type	Required
date	string (ISO date)	Yes
avg_temp_c	number	No
aqi	number	No
humidity_pct	number	No
precipitation_mm	number	No
source	string	No

ClimateMetricResponse

Field	Type	Required
id	integer	Yes
city_id	integer	Yes
date	string (ISO date)	Yes
avg_temp_c	number	Yes
aqi	number	Yes
humidity_pct	number	Yes
precipitation_mm	number	Yes
source	string	Yes
created_at	string (datetime)	No

NarrativeResponse

Field	Type	Required
`city_id`	integer	Yes
`city_name`	string	Yes
`narrative`	string	Yes

ObservationCreate

Field	Type	Required
`category`	string	Yes
`value`	number	Yes
`note`	string	No

ObservationResponse

Field	Type	Required
`id`	integer	Yes
`city_id`	integer	Yes
`user_id`	integer	Yes
`category`	string	Yes
`value`	number	Yes
`note`	string	Yes
`recorded_at`	string (datetime)	No
`created_at`	string (datetime)	No

ObservationUpdate

Field	Type	Required
`category`	string	No
`value`	number	No
`note`	string	No

SocioeconomicMetricCreate

Field	Type	Required
`year`	integer	Yes
`median_rent_gbp`	number	No
`green_space_pct`	number	No
`crime_index`	number	No
`avg_commute_min`	number	No

Field	Type	Required
`source`	string	No

SocioeconomicMetricResponse

Field	Type	Required
`id`	integer	Yes
`city_id`	integer	Yes
`year`	integer	Yes
`median_rent_gbp`	number	Yes
`green_space_pct`	number	Yes
`crime_index`	number	Yes
`avg_commute_min`	number	Yes
`source`	string	Yes
`created_at`	string (datetime)	No

TokenResponse

Field	Type	Required
`access_token`	string	Yes
`token_type`	string	No

UserLogin

Field	Type	Required
`username`	string	Yes
`password`	string	Yes

UserRegister

Field	Type	Required
`username`	string	Yes
`email`	string	Yes
`password`	string	Yes

UserResponse

Field	Type	Required
`id`	integer	Yes
`username`	string	Yes
`email`	string	Yes
`role`	string	Yes
`created_at`	string (datetime)	No

PaginatedResponse

All list endpoints return a paginated envelope:

Field	Type	Description
`items`	array	The result objects
`total`	integer	Total matching records
`offset`	integer	Current page offset
`limit`	integer	Page size

Error Response

Field	Type	Description
`detail`	string	Human-readable error message
`code`	string	Machine-readable error code

ValidationError (422)

Field	Type	Required
`loc`	array	Field location path
`msg`	string	Error description
`type`	string	Error type identifier